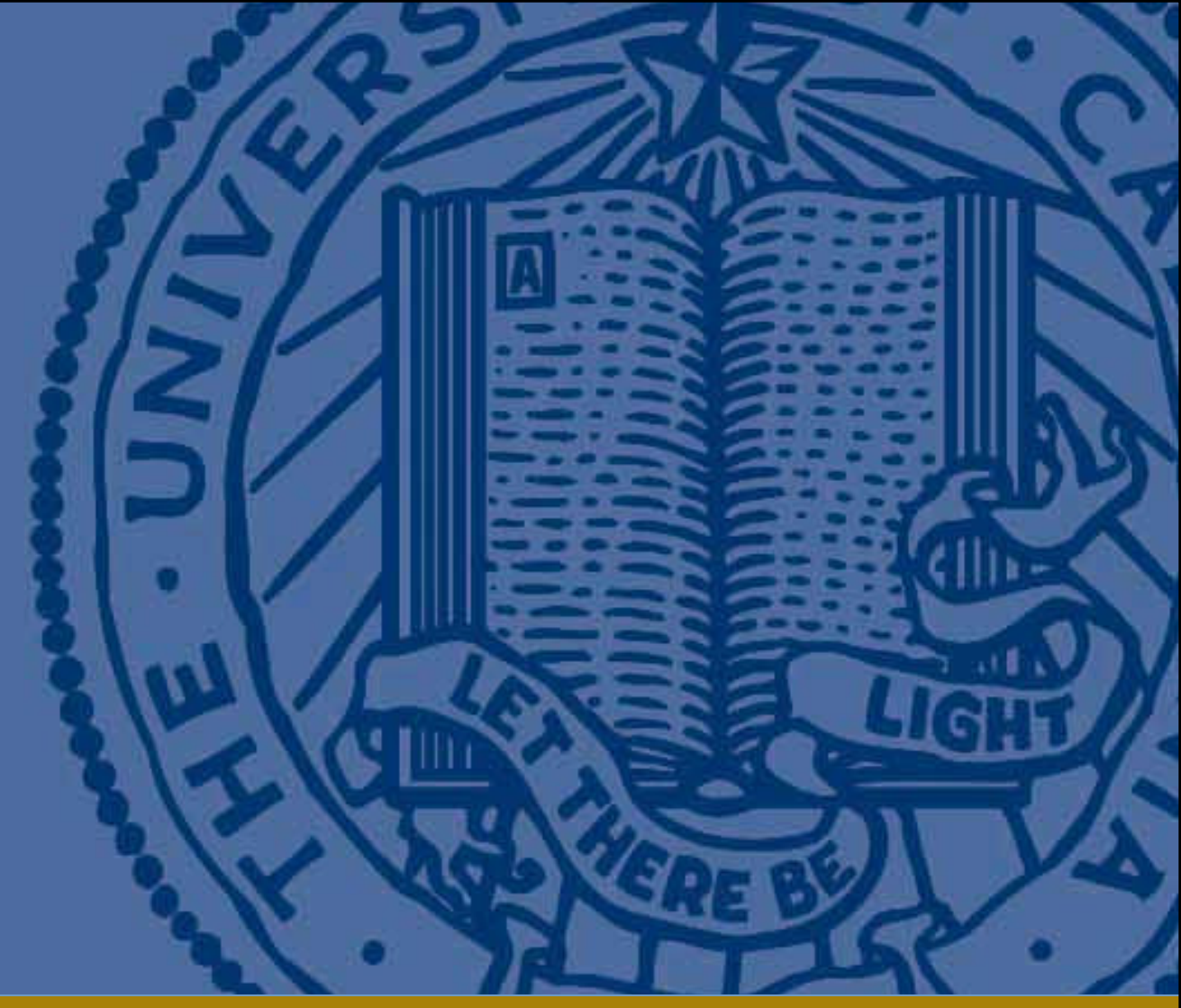


Structured Discourse Reference to Individuals



Adrian Brasoveanu

abrsvn@gmail.com

University of California at Santa Cruz

I. Multiple Donkey Anaphora

Main goal: argue that anaphora, i.e. discourse reference, in natural language involves two equally important components with essentially the same interpretive dynamics:

- reference to **values**: sets of objects, e.g. individuals, events, times, propositions etc.
- reference to **structure**, i.e. the correlation/dependency between values that is introduced and incrementally elaborated upon in discourse.

Benefit: structured discourse reference enables us to provide a precise compositional interpretation procedure for discourses involving complex descriptions of multiple related objects.

For example, consider the following two donkey sentences:

- Every^u person who buys a^{u'} book on amazon.com and has a^{u''} credit card uses it_{u''} to pay for it_{u'}.
(showing that singular donkey anaphora refers to non-singleton sets of individuals, i.e. non-singleton sets of values)
- Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.
(showing that singular donkey anaphora refers to a dependency between sets of values, i.e. structure)

Indexation convention: superscripts – new discourse referents are introduced; subscripts – previously introduced discourse referents are retrieved. A $\langle \text{superscript}, \text{subscript} \rangle$ pair is a co-reference arrow.

Sentence (1) – a **mixed maximal & non-maximal** donkey sentence (a.k.a. mixed strong & weak donkey sentence): for **every** book (maximal) that any credit card owner buys on amazon.com, there is **some** credit card (non-maximal) that s/he uses to pay for the book.

("uses a credit card" as opposed to a check etc.)

The credit card can vary from book to book, e.g. I can use my MasterCard to buy set theory books and my Visa to buy detective novels. That is: even the non-maximal indefinite a^{u''} credit card introduces a non-singleton set.

Sentence (2) – **two** instances of **maximal** donkey anaphora:

we are considering **every** Christmas gift and **every** girl;

the restrictor of the quantification introduces a dependency between the set of gifts and the set of girls:

each gift is correlated with **the** girl it was bought for;

the nuclear scope retrieves not only the sets of values, but also the structure associated with them, i.e. the dependency between them: **each** gift was wrapped by the deskmate of **the** girl that the gift was bought for;

we have here **donkey anaphora to structure** in addition to donkey anaphora to values.

II. Plural Compositional DRT (PCDRT)

This is the translation of (1) in classical first-order logic:

$$\begin{aligned} \forall x(\text{pers}(x) \wedge \exists y(\text{bk}(y) \wedge \text{buy}(x,y)) \wedge \exists z(\text{card}(z) \wedge \text{hv}(x,z)) \\ \rightarrow \forall y'(\text{bk}(y') \wedge \text{buy}(x,y')) \\ \rightarrow \exists z'(\text{card}(z') \wedge \text{hv}(x,z') \wedge \text{use_to_pay}(x,z',y')) \end{aligned}$$

The problem posed by sentence (1) – compositionally derive its interpretation while allowing for:

- the fact that the two donkey indefinites in the restrictor of the quantification receive two distinct readings
- the fact that the values of the non-maximal indefinite a^{u''} credit card co-vary with the values of the maximal indefinite a^{u'} book – although a^{u'} book cannot syntactically scope over a^{u''} credit card, since both DP's are trapped in their respective VP-conjuncts.

The main innovation relative to Muskens' Compositional DRT (CDRT) is that **dynamic information states are plural** (as in van den Berg's Dynamic Plural Logic):

- info states are modeled as **sets of sequences** of individuals, i.e. sets of variable assignments

Plural Info State J	...	u	u'	...
i ₁	...	x ₁ (=sam)	y ₁ (=red_dwarf)	...
i ₂	...	x ₂ (=megan)	y ₂ (=the_joke)	...
i ₃	...	x ₃ (=megan)	y ₃ (=life_is_elsewhere)	...
...	...	...	...	...

Values: the sets {x₁, x₂, x₃, ...} and {y₁, y₂, y₃, ...}

Structure: the relation {<x₁, y₁>, <x₂, y₂>, <x₃, y₃>, ...}

- DRS's **non-deterministically** update plural info states, i.e. they are **relations** between an input info state I and an output info state J:

$$[\text{new dref's} \mid \text{conditions}] := \lambda I. \lambda J. [I \text{ new dref's } J \wedge \text{conditions}]$$

A DRS constrains the output info state J in two ways: J differs from the input info state I at most with respect to the **new dref's** and J satisfies all the **conditions**.

E.g. *Some^u students read some^{u'} books:*

$$[u, u' \mid \text{student}\{u\}, \text{book}\{u'\}, \text{read}\{u, u'\}] := \lambda I. \lambda J. [I, u, u' J \wedge \text{student}\{u\} J \wedge \text{book}\{u'\} J \wedge \text{read}\{u, u'\} J]$$

(In the output info state J above, Sam read "Red Dwarf" and Megan read "The Joke" and "Life is Elsewhere")

Compositionality: the underlying logic is classical type logic, so compositionality at sub-clausal level follows automatically and standard techniques from Montague semantics (e.g. type shifting) become available.

III. Multiple Donkey Anaphora in PCDRT

Indefinites: non-deterministically introduce both values and structure, i.e. they introduce **structured sets of individuals**. The **maximal** / **non-maximal** donkey ambiguity is attributed to the indefinite articles.

(Attributing the ambiguity to the indefinites enables us to give a compositional account of mixed reading sentences like (1): we locally decide for each indefinite which reading it receives.)

maximal (strong): $a^{\text{str}:u} \text{book} \rightsquigarrow \lambda P. \max^u([\text{book}\{u\}]; P(u))$

non-maximal (weak): $a^{\text{wk}:u} \text{card} \rightsquigarrow \lambda P. [u \mid \text{card}\{u\}]; P(u)$

$$\max^u(D) := \lambda I. \lambda J. ([u]; D) / J \wedge \forall K([u]; D) / K \rightarrow uK \subseteq uJ, \text{ where } D \text{ is a DRS.}$$

(The **max** operator ensures that a strong indefinite introduces its maximal witness set, i.e. all individuals satisfying its restrictor and nuclear scope. In contrast, a weak indefinite introduces some – possibly non-maximal – witness set.)

Pronouns: anaphoric to the structured sets of individuals introduced by the indefinites, e.g. $it_{u'} \rightsquigarrow \lambda P. P(u')$

Selective generalized quantification – for simplicity, modeled as a test: $det^u \rightsquigarrow \lambda P. \lambda P'. [\det_u(P(u), P'(u))]$

$$\det_u(D, D') := \lambda I. I \neq \emptyset \wedge \text{DET}(u[D], u[D'; D'])$$

where D and D' are DRS's, **DET** is the corresponding static det., $u[D] := \cup \{uJ \mid ([u] \mid \text{unique}\{u\}); D) / J\}$ and $\text{unique}\{u\} := \lambda I. I \neq \emptyset \wedge \forall i \in I \forall i' \in I (u'i = u''i' \rightarrow u''i = u''i')$.

The **PCDRT representations** for sentences (1) and (2):

$$[\text{every}_u([\text{pers}\{u\}]; \max^u([\text{bk}\{u\}, \text{buy}\{u, u'\}]; [u'' \mid \text{card}\{u''\}, \text{hv}\{u, u''\}], [\text{use_to_pay}\{u, u', u''\}])]$$

$$[\text{every}_u([\text{boy}\{u\}]; \max^u([\text{gift}\{u\}]; \max^u([\text{girl}\{u'\}, \text{buy_for}\{u, u', u''\}]), \max^u([\text{deskmate}\{u''\}, \text{of}\{u'', u''\}]; [\text{unique}_{u''}\{u''\}]; [\text{ask_to_wrap}\{u, u'', u''\}])]$$

where $\text{unique}_{u''}\{u''\} := \lambda I. I \neq \emptyset \wedge \forall i \in I \forall i' \in I (u''i = u'''i' \rightarrow u'''i = u'''i')$.

$$[\text{pers}\{u\}];$$

for a person u
Sequence i ₀

$$\max^u([\text{bk}\{u\}, \text{buy}\{u, u'\}];$$

for a person u	store in u' all the books u bought
Sequence i ₁	
Sequence i ₂	
Sequence i ₃	

$$[u'' \mid \text{card}\{u''\}, \text{hv}\{u, u''\}]]$$

for a person u	store in u' all the books u bought	store in u'' some of u's cards
Sequence i ₁		
Sequence i ₂		
Sequence i ₃		

$$[\text{use_to_pay}\{u, u', u''\}]]$$

for each row in the matrix (i.e. each sequence), check that u paid for the u'-book in that row with the u''-card in that row
Sequence i ₁
Sequence i ₂
Sequence i ₃

IV. Generalizing Structured Reference

If we allow for **non-atomic (sum / plural) individuals** in our domain, we capture the fact that sentence (2) and sentence (3) below are intuitively parallel.

- Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

	u	u'	u''
Sequence i ₁			
Sequence i ₂			

- Every^u parent who gives a^{u'} balloon / three^{u'} balloons to two^{u''} boys expects them_{u''} to end up fighting (each other) for it_{u'} / them_{u''}.

	u	u'	u''
Sequence i ₁			
Sequence i ₂			

Also, PCDRT can be straightforwardly generalized to capture temporal anaphora, event anaphora, modal anaphora etc.

For example, we can capture the interaction between **anaphora to individuals**, **modal anaphora** and the entailment particle **therefore** in the following naturally-occurring discourse attributed to Thomas Aquinas,

- [A] man cannot live without joy. Therefore, when he is deprived of true spiritual joys, it is necessary that he become addicted to carnal pleasures.

which is intuitively interpreted as shown in (5) below.

- If a man is alive, he must find something pleasurable / he must have a pleasure.
- Therefore, if a man is alive and he does not have a spiritual pleasure, he must have a carnal pleasure.

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